Explanations for Independent Practice (Part 1 Only)

Topic 1 — Written Explanation Companion

This companion restates each Part 1 question, shows the checked answer, and explains the work very slowly. Follow the steps in order. Read every Important words list before you start the numbered steps.

How to use this companion (read once)

Instructor support, online resources, and AI are allowed on **explanations only**. On Independent Practice Parts 1 and 2, do **not** use the internet or AI. You may get **limited in-person help** only. These explanations prepare you for Part 1. If you can do Part 1 on your own, you can do Part 2. When you miss a problem, **circle** it, get help, then **rework** until you can solve it independently and correctly.

Lesson 1-1: Relate Integers and Opposites

Question 1

Question

A submarine rises 18 m from a point below sea level, then descends 18 m.
Part A: Represent both changes with signed numbers and find their sum.
Part B: Explain why the two changes are additive inverses.
Checked answer: $+18 + (-18) = 0$; equal magnitude and opposite signs.

What the question is asking

This question asks you to write each vertical change as a signed number — one for the rise and one for the descent — then add those two signed numbers to find their sum, and finally say why the two changes are additive inverses of each other.

Important words

- *Nonzero* means not equal to zero. The symbol $\neq$ means "is not equal to."
- A *fraction* $\frac{a}{b}$ means a equal parts out of b equal parts ($b \neq 0$).
- *Tenths* are equal parts made when one whole is split into 10 parts. *Hundredths* are equal parts made when one whole is split into 100 parts.
- A *decimal* uses a decimal point to show parts of a whole (tenths, hundredths, . . .).
- A *whole number* is $0, 1, 2, 3, \ldots$. It has no fraction or decimal part.
- A *terminating* decimal stops, like 0.5 or 2.25.
- A *repeating* decimal goes forever in a pattern, like $0.333\ldots$.
- An *integer* is one of the numbers $\ldots, -2, -1, 0, 1, 2, \ldots$; it has no decimal part.
- A *rational number* can be written as a fraction $\frac{a}{b}$ (with $b \neq 0$) or as a terminating or repeating decimal. The integer 18 is rational because $18 = \frac{18}{1}$.

- A *positive number* is greater than zero. A *negative number* is less than zero.
- A *signed* number (or signed move) uses a plus or minus sign to show direction: $+$ means positive (up / right / gain), and $-$ means negative (down / left / loss).
- To *cancel* (here) means two moves or numbers undo each other completely and leave zero.
- An *additive inverse* of a number is the number that has the **same size** but the **opposite direction**, so the two numbers add to zero (they cancel).
- *Vertical* means up and down.
- *Sea level* is the height we call zero. Below sea level is negative; above is positive.
- A *rise* of 18 m means moving **up** 18 meters, which we write as $+18$. A *descent* of 18 m means moving **down** 18 meters, which we write as -18.
- The *sum* is the answer we get when we add numbers together.
- *Downward* means the opposite direction of up.
- To *flip* the direction means change up$\leftrightarrow$down (or $+ \leftrightarrow -$) while keeping the same size.

Work the problem one tiny step at a time

1. **Part A:** Read the first change. The submarine **rises** 18 m.
2. Up is the positive direction, so write the rise as the signed number $+18$.
3. Read the second change. The submarine **descends** 18 m.
4. Down is the negative direction, so write the descent as the signed number -18.
5. Part A also asks for their sum, so add the two signed numbers: $+18 + (-18)$.
6. Adding -18 is the same as subtracting 18: $18 - 18 = 0$.
7. So the sum is 0. Write it as $+18 + (-18) = 0$.
8. **Part B:** Compare the two changes. Both have the **same size**: 18 meters.
9. Their directions are **opposite**: one is up ($+$) and one is down ($-$), so the signs are opposite.
10. Two numbers with the same size and opposite signs add to zero, which means they cancel. That is exactly what *additive inverse* means, so $+18$ and -18 are additive inverses.
11. Check: the submarine ends at the same depth it started, so the total change is 0 meters.

Final answer: Part A: $+18 + (-18) = 0$; Part B: the two changes have equal size and opposite signs, so they are additive inverses.

Why this makes sense

The rise and the descent cover the same distance in opposite directions, so the descent undoes the rise exactly. The submarine finishes at the depth it started from, and the two signed numbers cancel to zero.

Question 2

Question

On a number line, zero is located in the middle. We have two mystery numbers, a and b. We are told that $a < 0$, $b > 0$, and $a + b = 0$. Write an equation about the absolute values of a and b that must be true.
Checked answer: $|a| = |b|$

What the question is asking

This question asks for a math sentence with an equals sign that compares how far a and b are from zero (distance from zero, which is never negative).

Important words

- *Negative* means less than zero (left of zero). The symbol $<$ means "less than."
- *Positive* means greater than zero (right of zero). The symbol $>$ means "greater than."
- A *number line* is a straight line with numbers in order. Zero is in the middle. Negative numbers are left of zero. Positive numbers are right of zero.
- A *variable* is a letter that stands for a number. Here a and b are variables.
- An *equation* is a math sentence with an equals sign that says two sides are equal.
- An *additive inverse* of a number is the number with the same size but the opposite direction, so the two numbers add to zero.
- To *cancel* (here) means two numbers undo each other completely and leave zero (they are additive inverses).
- *Distance* tells how far apart two points are. Distance is never negative.
- *Absolute value* of a number, written $|\ |$, is the distance from that number to zero. Distance is never negative.

Work the problem one tiny step at a time

1. We are told $a + b = 0$.
2. That means b is the additive inverse of a: $b = -a$.
3. So a and b sit on **opposite sides** of zero.
4. Because they cancel (add to zero), they must be the **same distance** from zero.
5. Distance from zero is absolute value. So $|a|$ equals $|b|$.
6. Write the equation: $|a| = |b|$.
7. Numeric check: if $a = -5$ and $b = 5$, then $a + b = 0$, and $|-5| = 5 = |5|$.

Final answer: $|a| = |b|$

Why this makes sense

If two numbers add to zero, they are additive inverses (opposites). Opposites are always the same distance from zero, so their absolute values match. A common mistake is writing $a = b$ (false, because one is negative and one is positive). Check: $-5 + 5 = 0$ and $|-5| = |5|$.

Question 3

Question

Evaluate: $|-9| =$
Checked answer: 9

What the question is asking

This question asks how far -9 is from zero on a straight line of numbers (distance from zero is never negative).

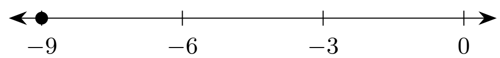
Important words

- An *operation* is a math action. Add joins amounts. Subtract takes away or finds a difference. Multiply makes equal groups. Divide splits into equal groups.
- An *expression* is a math phrase made of numbers, operations, and sometimes symbols like absolute-value bars (it does not have to be a full equation).
- A *decimal* uses a decimal point to show parts of a whole.
- To *evaluate* means to find the value of an expression.
- A *negative number* is less than zero. A *positive number* is greater than zero.
- A *number line* is a straight line with numbers in order. Zero is in the middle. Negatives are left; positives are right.
- *Distance* tells how far apart two points are. Distance is never negative.
- *Absolute value* $|\ |$ means distance from zero on the number line. Distance is never negative.
- An *integer* is one of the numbers $\ldots, -2, -1, 0, 1, 2, \ldots$; it has no decimal part.
- A *unit* on the number line is one tick-mark step from one integer to the next.
- The number -9 is nine units left of zero.
- The absolute-value bars do **not** mean "make the number negative." They report how far the number sits from zero.

Work the problem one tiny step at a time

1. Picture a straight line of numbers. Mark zero in the middle.
2. Find -9. It is 9 steps to the left of zero.
3. Count the steps from 0 to -9 one by one: $1, 2, 3, 4, 5, 6, 7, 8, 9$. That is 9 steps.
4. Distance does not care about left or right. Keep the count only: 9.

5. Absolute value writes that distance with bars: $|-9| = 9$.
6. Check another way: absolute value of a negative number is the positive version of that number's size, so $|-9|$ matches the size 9.
7. Contrast check: $|9| = 9$ also (same distance on the right side). Distance from zero is 9 either way.



Final answer: 9

Why this makes sense

Absolute value never gives a negative answer. The bars turn -9 into the positive distance 9. Check: nine steps left of zero is still a distance of 9. A common mistake is thinking $|-9| = -9$.

Question 4

Question

Order from least to greatest: $-3,\ 8,\ -1$
Checked answer: $-3,\ -1,\ 8$

What the question is asking

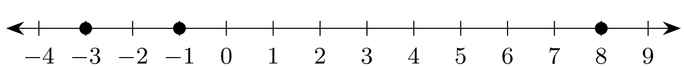
This question asks you to list the three numbers from smallest to largest.

Important words

- *Least to greatest* means smallest first, then bigger, then biggest.
- On a *number line*, numbers farther to the **left** are smaller. Numbers farther to the **right** are larger.
- A *negative* number is less than zero. A *positive* number is greater than zero.
- *Absolute value* means distance from zero. Ordering by absolute value alone is wrong because it ignores which side of zero a number is on.

Work the problem one tiny step at a time

1. Draw a number line and mark -3, -1, and 8.
2. Look which mark is farthest left: -3. That is the least.
3. Next from the left: -1.
4. Farthest right: 8. That is the greatest.
5. Write them in that order: $-3,\ -1,\ 8$.



Final answer: $-3,\ -1,\ 8$

Why this makes sense

Left means smaller. -3 is left of -1, and both negatives are left of positive 8. A common mistake is ordering by absolute value and writing $-1, -3, 8$.

Lesson 1-2: Understand Rational Numbers

Question 1

Question

In a survey, 3 out of 8 students chose basketball as their favorite sport. What is the decimal equivalent of the fraction of students who chose basketball?
Checked answer: 0.375

What the question is asking

This question asks you to change $\frac{3}{8}$ into a number that uses a dot for parts of a whole.

Important words

- A *fraction* $\frac{3}{8}$ means 3 parts out of 8 equal parts. The top number is the *numerator*. The bottom number is the *denominator*.
- A *decimal* uses a decimal point to show parts of a whole, like 0.375.
- *Equivalent* means "equal in value."
- *Long division* is a written way to divide one digit at a time and keep track of what is left.
- A fraction bar means *divide*: top $\div$ bottom.
- A *remainder* is what is left over after a division step.
- To *bring down* a digit means move the next digit (often a 0) into the current remainder so you can keep dividing.
- A *decimal digit* is a digit to the right of the decimal point (after the dot).

Work the problem one tiny step at a time

1. Write the fraction: $\frac{3}{8}$.
2. Divide: $3 \div 8$. Since 3 is smaller than 8, write 3.000.
3. 8 goes into 30 three times: $8 \times 3 = 24$. Subtract: $30 - 24 = 6$. Remainder 6.
4. Bring down 0 to make 60. 8 goes into 60 seven times: $8 \times 7 = 56$. Subtract: $60 - 56 = 4$. Remainder 4.
5. Bring down 0 to make 40. 8 goes into 40 five times: $8 \times 5 = 40$. Subtract: $40 - 40 = 0$. Remainder 0.
6. The decimal digits are 3, then 7, then 5. So $3 \div 8 = 0.375$.

Final answer: 0.375

Why this makes sense

Long division stops with remainder 0, so the decimal terminates (stops; no leftover repeating pattern) at 0.375. Check: $0.375 \times 8 = 3$.

Question 2

Question

A student uses long division to convert a fraction $\dfrac{x}{y}$, where $y \neq 0$, into a decimal. Which outcome is **NOT** possible?

A) The decimal terminates (stops).

B) The decimal repeats a pattern.

C) The decimal neither terminates nor repeats.

Checked answer: C

What the question is asking

This question asks which outcome cannot happen when you divide the numerator of a fraction by its nonzero denominator.

Important words

- *Long division* is a written way to divide one digit at a time and keep track of what is left.
- A *fraction* $\frac{x}{y}$ means the top number x is divided by the bottom number y.
- The *numerator* is the top number being divided. Here it is x.
- The *denominator* is the bottom number that divides the top number. Here it is y.
- *Nonzero* means not equal to zero. The denominator must be nonzero because division by zero is not allowed.
- A *whole number* is one of the numbers $0, 1, 2, 3, \ldots$.
- A *decimal* uses a decimal point to show parts smaller than one whole.
- A *terminating decimal* stops, like 0.5 or 0.375.
- A *repeating decimal* goes on forever but in a repeating pattern, like $0.333\ldots$.
- A *quotient* is the answer to a division problem.

Work the problem one tiny step at a time

1. When you divide two whole numbers (nonzero denominator), the decimal **must** either terminate or repeat.
2. Statement A can be true for some fractions (example: $\frac{1}{2} = 0.5$).
3. Statement B can be true for some fractions (example: $\frac{1}{3} = 0.333\ldots$).

4. Statement C says the decimal neither terminates nor repeats. That outcome is **not possible** for a fraction of whole numbers.

5. So the outcome that is NOT possible is C.

Final answer: C

Why this makes sense

Fractions of whole numbers always become terminating or repeating decimals. Statement C describes something that does not happen for those fractions.

Question 3

Question

In a class debate, Jordan says $0.\overline{27}$ is rational because its digits repeat. Casey says it is not rational because it never ends. Which student is correct? Use a fraction or equation to support the claim.
Checked answer: Jordan; $0.\overline{27} = \dfrac{3}{11}$.

What the question is asking

Decide whether a repeating decimal is rational and prove it.

Important words

- A repeating decimal has a digit pattern that repeats forever.
- A rational number can be written as a fraction of integers with a nonzero denominator.

Work the problem one tiny step at a time

1. Let $x = 0.\overline{27}$.
2. Multiply by 100: $100x = 27.\overline{27}$.
3. Subtract: $99x = 27$, so $x = \dfrac{27}{99} = \dfrac{3}{11}$.

Final answer: Jordan is correct; $0.\overline{27} = \dfrac{3}{11}$.

Why this makes sense

The decimal has an exact fraction name, so it is rational.

Lesson SC-1: Compare Rational Numbers

Question 1

Question

Your pet cat weighs 10.4 pounds. Your friend's rabbit weighs 10.6 pounds. Write two different statements using the less than ($<$) and greater than ($>$) symbols that compare the weight of the cat and the rabbit.
Checked answer: $10.4 < 10.6$ and $10.6 > 10.4$

What the question is asking

This question asks for two true comparison sentences relating 10.4 and 10.6 using the less-than mark ($<$) and the greater-than mark ($>$).

Important words

- A *decimal* has digits after the decimal point. Here both weights have one digit after the point.
- The *ones place* is the place just left of the decimal point. It counts whole ones.
- *Tenths* is the first digit place to the right of the decimal point.
- *Hundredths* is the second digit place to the right of the decimal point.
- *Place value* is the value of a digit's position (ones, tenths, hundredths, ...).
- The symbol $<$ means *less than* (smaller).
- The symbol $>$ means *greater than* (larger).
- To *compare* means to decide which is smaller or larger.
- To *flip* a comparison means write the same fact the other way ($a < b$ becomes $b > a$).

Work the problem one tiny step at a time

1. Line up the decimals by place value: both start with 10.
2. Compare the tenths place: 4 tenths versus 6 tenths.
3. Because $4 < 6$, we have $10.4 < 10.6$.
4. Flip the comparison the other way: $10.6 > 10.4$.
5. Both statements are true and use different symbols as required.

Final answer: $10.4 < 10.6$ and $10.6 > 10.4$ (pounds OK)

Why this makes sense

The cat is lighter. Saying the smaller number is less than the larger number, and the larger is greater than the smaller, are two ways to say the same fact.

Question 2

Question

Compare the fraction $\frac{3}{4}$ with the decimal 0.75. Select all symbols that appropriately relate the two numbers.

$$< \qquad > \qquad = \qquad \neq$$

Checked answer: $=$ (only)

What the question is asking

This question asks which comparison symbols are true for $\frac{3}{4}$ and 0.75.

Important words

- A *fraction* $\frac{3}{4}$ means 3 divided by 4.
- A *decimal* is a number written with a decimal point to show parts smaller than one whole. Here the decimal is 0.75.
- *Long division* is a written way to divide one digit at a time.
- A *remainder* is the amount left after one division step.
- To *bring down* a digit means move the next digit beside the remainder so division can continue.
- The symbol $=$ means *equal to.*
- The symbol $<$ means *less than* (strictly smaller, never equal).
- The symbol $>$ means *greater than* (strictly larger, never equal).
- The symbol $\neq$ means *not equal to.*
- *Select all* means check every symbol on the list and keep only the ones that make a true statement. Here the list offers just four symbols: $<$, $>$, $=$, and $\neq$.

Work the problem one tiny step at a time

1. Change the fraction to a decimal: $\frac{3}{4} = 3 \div 4$.
2. Write 3 as 3.00 so the division can continue past the decimal point.
3. Bring down the first 0. This makes 30.
4. The number 4 fits into 30 seven times.
5. Multiply to check that digit: $4 \times 7 = 28$.
6. Subtract: $30 - 28 = 2$. The remainder is 2.
7. Bring down the next 0. This makes 20.
8. The number 4 fits into 20 five times because $4 \times 5 = 20$.
9. Subtract: $20 - 20 = 0$. The remainder is 0.
10. The decimal digits are 7 and 5. So $3 \div 4 = 0.75$.
11. Compare the two numbers: $\frac{3}{4} = 0.75$ exactly. So $=$ is true.
12. Now test each of the other three symbols offered.
13. Test $<$: it would say $\frac{3}{4}$ is strictly smaller than 0.75. The numbers are equal, so $<$ is false.
14. Test $>$: it would say $\frac{3}{4}$ is strictly larger than 0.75. The numbers are equal, so $>$ is false.
15. Test $\neq$: it would say the two numbers are not equal. They are equal, so $\neq$ is false.
16. Only one symbol out of the four makes a true statement, so select $=$ and nothing else.

Final answer: = (only)

Why this makes sense

The fraction $\frac{3}{4}$ and the decimal 0.75 are two ways of writing the very same number, so the only relationship that holds between them is equality. Each of the other three choices claims the numbers differ in some way — smaller, larger, or unequal — and every one of those claims is false when the numbers are the same.

Lesson 1-3: Add Integers

Question 1

Question

In the first round of a board game, a player scores 10 points. On their next turn, they lose 4 points. On their last turn of the round, they lose 8 points.
Part A: Explain how to show the change in score using a number line.
Part B: What is the final score?
Checked answer: Part A: start at 0, move right 10, left 4, left 8; Part B: -2

What the question is asking

This question asks you to show the score changes on a straight line of numbers and find the final score after $+10$, then -4, then -8.

Important words

- A *decimal* uses a decimal point to show parts of a whole.
- An *integer* is one of the numbers $\ldots, -2, -1, 0, 1, 2, \ldots$; it has no decimal part.
- A *positive number* is greater than zero. A *negative number* is less than zero.
- To *add* means join an amount to the amount you have.
- A *number line* is a straight line with numbers in order. Moving right adds a positive number. Moving left adds a negative number.
- A *unit* on the number line is one tick-mark step from one integer to the next; on this number line one unit is one point.
- A *score* of 10 points is $+10$. Losing 4 points is -4. Losing 8 points is -8.
- The *final score* is what you get after all the moves.

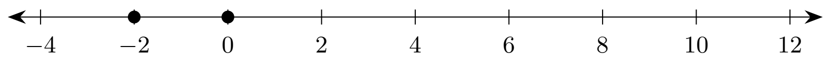
Work the problem one tiny step at a time

1. **Part A:** Start at 0 on a number line.
2. Move right 10 units to land on 10 (score 10 points).
3. Move left 4 units to land on 6 (lose 4).
4. Move left 8 units to land on -2 (lose 8).

5. **Part B:** Add the integers: $10 + (-4) + (-8)$.

6. First: $10 + (-4) = 6$.

7. Then: $6 + (-8) = -2$.



Final answer: Part A: right 10, left 4, left 8; Part B: -2

Why this makes sense

Gains go right; losses go left. Ending left of zero means a negative score. Check: $10 - 4 - 8 = -2$.

Question 2

Question

A student starts a game with 15 points. They lose 20 points in the first round and then gain 3 points in the second round. What is the student's final score?
Checked answer: -2

What the question is asking

This question asks for the final score after starting at 15, losing 20, then gaining 3.

Important words

- An *operation* is a math action. Add joins amounts. Subtract takes away or finds a difference. Multiply makes equal groups. Divide splits into equal groups.
- An *expression* is a math phrase made of numbers and operation signs.
- A *decimal* uses a decimal point to show parts of a whole.
- An *integer* is one of the numbers $\ldots, -2, -1, 0, 1, 2, \ldots$; it has no decimal part.
- To *lose* points means add a *negative* integer.
- To *gain* points means add a *positive* integer.
- On a *number line,* moving left adds a negative; moving right adds a positive.
- The *final score* is the value after all the additions are done.
- An *addend* is a number being added. Here the addends are 15, -20, and 3.

Work the problem one tiny step at a time

1. Write the expression: $15 + (-20) + 3$.
2. First combine $15 + (-20)$. Start at 15. Move left 20.
3. From 15, move left 15 units to reach 0. That uses 15 of the 20 leftward steps.
4. Steps still left: $20 - 15 = 5$. From 0, move left 5 more to land on -5.

5. So $15 + (-20) = -5$.
6. Now add 3: $-5 + 3$.
7. Start at -5. Move right 3. Land on -2.
8. So the final score is -2.
9. Check: $15 - 20 = -5$, then $-5 + 3 = -2$. Same answer.

Final answer: -2

Why this makes sense

Losing more than you started with puts you below zero, then gaining 3 only climbs back a little: $-5 + 3 = -2$. Check: $15 - 20 + 3 = -2$.

Question 3

Question

A bank account starts with \$500. The bank deducts a monthly maintenance fee of \$12.50. If there are no other deposits or withdrawals, how much money is in the account after 6 months?
Checked answer: \$425

What the question is asking

This question asks how much money is left after six monthly charges of \$12.50.

Important words

- A *balance* is how much money is in the account.
- A *negative* number is less than zero.
- To *multiply* means make equal groups. For example, $6 \times \$12.50$ means 6 equal groups of \$12.50.
- To *subtract* means take away or find what remains.
- A *fee* is money taken away. Taking away means subtract (or add a negative).
- A *deposit* adds money. A *withdrawal* removes money. Here there are none besides the fee.
- *Monthly* means once each month.

Work the problem one tiny step at a time

1. Fee each month: \$12.50.
2. Number of months: 6.
3. Total fees: multiply $6 \times \$12.50$.
4. Compute $6 \times \$12 = \$72$ and $6 \times \$0.50 = \$3$, then $\$72 + \$3 = \$75$. Total fees $= \$75$.
5. Subtract from the start: $\$500 - \$75$.
6. Compute $\$500 - \$75 = \$425$.

Final answer: \$425

Why this makes sense

Six fees of \$12.50 make \$75. Starting at \$500 and removing \$75 leaves \$425. A common mistake is subtracting \$12.50 only once.

Question 4

Question

Evaluate: $-12 + 19 =$
Checked answer: 7

What the question is asking

This question asks for the value of $-12 + 19$. You start left of zero and move right by a larger amount.

Important words

- An *operation* is a math action. Add joins amounts. Subtract takes away or finds a difference. Multiply makes equal groups. Divide splits into equal groups.
- An *expression* is a math phrase made of numbers and operation signs.
- To *evaluate* means find the value of an expression.
- A *unit* is one single step or amount.
- A *number line* puts numbers in order: negatives left of zero, positives right of zero.
- Adding a *positive* number on a number line means move **right**.
- -12 is 12 units **left** of zero. $+19$ means move 19 units right from wherever you start.
- *Absolute value* means distance from zero. Here $|-12| = 12$ and $|19| = 19$.
- An *addend* is a number being added. In $-12 + 19$, the addends are -12 and 19.
- The *sum* is the answer to an addition problem.
- When the positive addend is larger than the absolute value of the negative start, the sum is positive.

Work the problem one tiny step at a time

1. Start at -12 on the number line (12 left of zero).
2. Add 19 means move right 19 units from -12.
3. First, move right 12 units to reach 0. That uses up 12 of the 19 steps.
4. Steps used so far: 12. Steps still left: $19 - 12 = 7$.
5. From 0, move right the remaining 7 units.
6. Land on $+7$. So $-12 + 19 = 7$.

7. Check another way: the larger absolute value is 19 (positive), and $19 - 12 = 7$, so the sum is $+7$.

Final answer: 7

Why this makes sense

You had to "pay back" 12 just to get back to zero, then 7 steps remained to the right. Check: $19 - 12 = 7$, and the answer is positive because 19 is larger than 12.

Lesson 1-4: Subtract Integers

Question 1

Question

At 6:00 PM, the temperature was 5°F. By midnight, the temperature was -8°F. Write at least two mathematical expressions that correctly represent the distance between the two temperatures.
Checked answer: Distance 13°F; any correct distance expression such as $|5 - (-8)|$, $|-8 - 5|$, $5 - (-8)$, $|-8| + 5$

What the question is asking

This question asks for math phrases that show how far apart 5°Fand -8°Fare.

Important words

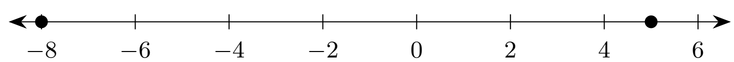
- An *operation* is a math action. Add joins amounts. Subtract takes away or finds a difference. Multiply makes equal groups. Divide splits into equal groups.
- An *expression* is a math phrase made of numbers and operation signs.
- *Distance* between two numbers is how far apart they are. Distance is never negative.
- A *difference* is the answer to a subtraction problem (here, one temperature minus the other).
- *Absolute value* $|\ |$ turns a difference into a positive distance.
- *Subtracting a negative*: $a - (-b) = a + b$.
- To *flip* the order means subtract the other way; absolute value keeps the distance positive.
- 8°Fbelow zero means -8°F.

Work the problem one tiny step at a time

1. The two temperatures are 5 and -8.
2. Distance $= |5 - (-8)|$.
3. Subtracting a negative changes to adding: $5 - (-8) = 5 + 8$.
4. Add: $5 + 8 = 13$.
5. Absolute value: $|13| = 13$.
6. Flip the subtraction order to make another expression: $|-8 - 5|$.

7. Work inside the bars: $-8 - 5 = -13$.
8. Take the absolute value: $|-13| = 13$.
9. The expression $5 - (-8)$ is also valid. Its value is the 13 found above.
10. Another expression is $|-8| + 5$.
11. First find the absolute value: $|-8| = 8$.
12. Then add: $8 + 5 = 13$.

Final answer: Distance 13°F; e.g. $|5 - (-8)|$, $|-8 - 5|$, $5 - (-8)$, $|-8| + 5$



Marked points: -8°F and 5°F; distance = 13.

Why this makes sense

From 5 down to 0 is 5 degrees, then down to -8 is 8 more, total 13. Check: $5 + 8 = 13$. Any expression that equals 13 works.

Question 2

Question

Two birds are flying in the sky. Bird A is 150 feet above sea level. Bird B is 120 feet above sea level. Write at least two expressions that represent the vertical distance between the birds.
Checked answer: Distance 30 ft; e.g. $|150 - 120|$, $|120 - 150|$, $150 - 120$

What the question is asking

This question asks for math phrases that show how far apart the birds are up and down.

Important words

- *Vertical* means up and down (not side to side).
- *Vertical distance* means the up-down distance between two heights.
- A *positive* number is greater than zero.
- A *negative* number is less than zero.
- *Above sea level* means a positive height.
- An *operation* is a math action. Add joins amounts. Subtract takes away or finds a difference. Multiply makes equal groups. Divide splits into equal groups.
- An *expression* is a math phrase made of numbers and operation signs that names a value.
- *Absolute value* keeps distance positive even if you subtract in either order.

Work the problem one tiny step at a time

1. Heights: Bird A is 150 ft. Bird B is 120 ft.
2. Both are above sea level, so both heights are positive.
3. Distance $= |150 - 120|$.
4. Compute $150 - 120 = 30$.
5. So $|150 - 120| = 30$.
6. Flipped order (subtract the other way; absolute value keeps distance positive): $|120 - 150| = |-30| = 30$.
7. Also: $150 - 120 = 30$ (already positive).

Final answer: Distance 30 ft; e.g. $|150 - 120|$, $|120 - 150|$, $150 - 120$

Why this makes sense

The higher bird is 30 feet above the lower bird. Absolute value protects you if you subtract the other way.

Question 3

Question

Two fish are swimming beneath the water. Fish A's elevation is -25 feet relative to sea level. Fish B's elevation is -40 feet relative to sea level. Write at least two expressions that represent the vertical distance between the fish.
Checked answer: Distance 15 ft; e.g. $|-25 - (-40)|$, $|-40 - (-25)|$, $40 - 25$

What the question is asking

This question asks for math phrases that show how far apart -25 ft and -40 ft are up and down.

Important words

- A *positive* number is greater than zero.
- A *negative* number is less than zero.
- *Below sea level* is a negative height.
- *Vertical distance* means how far apart two heights are up and down.
- A *signed number* has a plus or minus sign that shows direction.
- A *difference* is the answer to a subtraction problem.
- An *expression* is a math phrase that names a value.
- *Absolute value* $|\ |$ means distance from zero, so it is never negative.
- *Distance* is still positive. Use absolute value of the difference.
- Subtracting a negative: $-25 - (-40) = -25 + 40$.

Work the problem one tiny step at a time

1. Positions: Fish A at -25 ft. Fish B at -40 ft.
2. Both are below sea level, so both heights are negative.
3. Distance $= |-25 - (-40)|$.
4. Inside: $-25 - (-40) = -25 + 40 = 15$.
5. Absolute value: $|15| = 15$.
6. Flipped order (subtract the other way): $|-40 - (-25)| = |-40 + 25| = |-15| = 15$.
7. Using depths as positive amounts: deeper minus shallower $40 - 25 = 15$.

Final answer: Distance 15 ft; e.g. $|-25 - (-40)|$, $|-40 - (-25)|$, $40 - 25$

Why this makes sense

Fish B is 15 feet deeper than Fish A. Same distance whether you use signed heights or positive depths.

Question 4

Question

An elevator starts on floor -3, travels to floor 8, then returns to floor -1. What total distance does it travel?
Checked answer: 20 floors

What the question is asking

Find two whole-number distances across ground level and add them.

Important words

- *Positive floors* are above ground; *negative floors* are below ground.
- *Distance* is always positive.

Work the problem one tiny step at a time

1. From -3 to 8: $|8 - (-3)| = 11$.
2. From 8 to -1: $|-1 - 8| = 9$.
3. Add: $11 + 9 = 20$.

Final answer: 20 floors

Why this makes sense

Each leg covers a whole number of floors, so the two distances add to 20.

Lesson 1-5: Add and Subtract Rational Numbers

Question 1

Question

Another bank account starts with \$200. The bank charges a fee of \$15 each month, but *only* if the balance is below \$250. If there are no other deposits or withdrawals, what is the balance after 4 months?
Checked answer: \$140

What the question is asking

This question asks for the money left after 4 months when a \$15 fee applies only while the money in the account is below \$250.

Important words

- A *balance* is the money in the account.
- A *fee* subtracts money when the rule says so.
- *Below* \$250 means the balance is less than \$250.
- To *compare* means decide whether one number is less than, equal to, or greater than another.
- The symbol $<$ means "is less than."

Work the problem one tiny step at a time

1. Start: \$200. Compare to \$250: \$200 $<$ \$250, so the fee applies.
2. Month 1: \$200 $-$ \$15 $=$ \$185. Still \$185 $<$ \$250, fee will apply again.
3. Month 2: \$185 $-$ \$15 $=$ \$170. Still below \$250.
4. Month 3: \$170 $-$ \$15 $=$ \$155. Still below \$250.
5. Month 4: \$155 $-$ \$15 $=$ \$140. Still below \$250.
6. Check another way: 4 fees of \$15 is $4 \times$ \$15 $=$ \$60. Then \$200 $-$ \$60 $=$ \$140. Same answer.

Final answer: \$140

Why this makes sense

The account never reaches \$250, so the fee hits every month. Four fees remove \$60 from \$200, leaving \$140.

Question 2

Question

At 6:00 PM, the temperature is -3.75°C. It drops 2.4°C, then drops 1.85°C. What is the temperature at midnight?
Checked answer: -8°C

What the question is asking

Combine a negative starting temperature and two negative changes.

Important words

- A *drop* is a negative change.
- A *terminating decimal* ends.

Work the problem one tiny step at a time

1. $-3.75 - 2.4 - 1.85 = -8.00$.
2. Remove unnecessary zeros: -8.

Final answer: $-8°\mathrm{C}$

Why this makes sense

Both drops move the temperature farther below zero.

Question 3

Question

A lookout is $2\frac{1}{2}$ miles above sea level. A mountain base is 1.75 miles below sea level. What is the vertical distance?
Checked answer: 4.25 miles

What the question is asking

Find distance between a positive mixed number and a negative decimal.

Important words

- *Above sea level* is positive.
- *Below sea level* is negative.

Work the problem one tiny step at a time

1. $2\frac{1}{2} = 2.5$.
2. $|2.5 - (-1.75)| = 4.25$.

Final answer: 4.25 miles

Why this makes sense

The locations lie on opposite sides of sea level, so their distances from zero add.

Lesson 1-6: Multiply Integers

Before Questions 3 and 4, remember the order-of-operations rule called **PEMDAS**.

- **P** means *Parentheses*: do work inside ( ) first.
- **E** means *Exponents*: do powers like 5^2 next.
- **MD** means *Multiply* and *Divide*: left to right.
- **AS** means *Add* and *Subtract*: left to right last.

Question 1

Question

A hiker begins at sea level, 0 feet, and descends into a canyon at a steady rate of 15 feet per minute.
Part A: What integer represents the unit rate of their descent?
Part B: Where will they be in relation to sea level after 12 minutes?
Checked answer: Part A: -15; Part B: -180 feet (below sea level)

What the question is asking

This question asks for the up-or-down change for one minute of going down (down is negative), then the position after 12 minutes.

Important words

- *Sea level* is 0 feet.
- *Descending* means going down, so the change is *negative.*
- A *positive* number is greater than zero. A *negative* number is less than zero.
- A *unit* is one single amount. Here the time unit is one minute.
- A *unit rate* tells how much change happens in one unit of time (here, one minute).
- A *decimal* uses a decimal point to show parts of a whole.
- An *integer* is one of the numbers $\ldots, -2, -1, 0, 1, 2, \ldots$; it has no decimal part.
- To find total change, *multiply* rate by time.

Work the problem one tiny step at a time

1. **Part A:** Down 15 feet each minute means the rate is -15 feet per minute.
2. **Part B:** Multiply rate by time: $(-15) \times 12$.
3. Ignore signs first. Break 15×12: $15 \times 10 = 150$ and $15 \times 2 = 30$, then $150 + 30 = 180$.
4. Negative times positive is negative: $(-15) \times 12 = -180$.
5. -180 feet means 180 feet below sea level.

Final answer: Part A: -15; Part B: -180 feet (below sea level)

Why this makes sense

Steady descent stacks the same negative move 12 times. $12 \times 15 = 180$ feet down.

Question 2

Question

A remote control plane is descending at a rate of 250.50 meters per second. What is the change in the plane's height after $4\frac{1}{2}$ seconds? Round your answer to the nearest kilometer.
Checked answer: -1 kilometer

What the question is asking

This question asks for the up-or-down height change after 4.5 seconds of going down (down is negative), rounded to the nearest kilometer (a kilometer is 1000 meters).

Important words

- *Descending* means height change is negative.
- *Nonzero* means not equal to zero.
- A *unit* is one single amount. Here the time unit is one second.
- A *rate* tells how much change happens during one unit of time. Here the plane descends 250.50 meters during each 1 second.
- A *whole number* is $0, 1, 2, 3, \ldots$ with no fraction or decimal part.
- A *fraction* has a top number and a nonzero bottom number and names equal parts of a whole.
- A *mixed number* $4\frac{1}{2}$ means $4 + \frac{1}{2} = 4.5$.
- A *decimal* uses a dot called a *decimal point* to show parts of a whole. The *decimal part* is the part to the right of that dot.
- The *tenths digit* is the first digit to the right of the decimal point.
- A *product* is the answer to multiplication. A *quotient* is the answer to division.
- A *kilometer* is 1000 meters.
- To *round to the nearest kilometer,* look at the decimal part of the kilometer amount.

Work the problem one tiny step at a time

1. Convert time: $4\frac{1}{2} = 4.5$ seconds.
2. Multiply rate by time: 250.50×4.5.
3. First find 250.50×4. Multiply the whole-number part: $250 \times 4 = 1000$. Multiply the decimal part: $0.50 \times 4 = 2.00$. Add: $1000 + 2.00 = 1002.00$.
4. Then find 250.50×0.5. Multiplying by 0.5 means take one half. Half of 250 is 125. Half of 0.50 is 0.25. Add: $125 + 0.25 = 125.25$.
5. Add the two products. Line up the decimal points: $1002.00 + 125.25 = 1127.25$ meters down.
6. Change to kilometers by dividing by 1000. Move the decimal point left past three digits: $1127.25 \to 112.725 \to 11.2725 \to 1.12725$. So the change is 1.12725 km down.
7. Round to nearest km: look at the tenths digit 1 (less than 5), so round to 1 km down.
8. Write the signed change: -1 kilometer.

Final answer: -1 kilometer

Why this makes sense

About 1127 meters is a little more than 1 km, but closer to 1 km than to 2 km when rounding. Descent keeps the negative sign.

Question 3

Question

Evaluate: $(-5)^2 =$
Checked answer: 25

What the question is asking

This question asks for the value of $(-5)^2$. The curved marks wrap -5, then multiply that number by itself.

Important words

- *Parentheses* ( ) group a quantity. Here they wrap -5.
- A *signed number* has a plus or minus sign that shows direction.
- A *factor* is a number being multiplied. In $(-5) \times (-5)$, each -5 is a factor.
- A *base* is the number used as a repeated factor. Here the base is the full signed number -5.
- An *exponent* is the small raised number that tells how many times to use the base as a factor. In $(-5)^2$, the exponent is 2.
- A *power* is a base and its exponent together, such as $(-5)^2$.
- To *square* means raise to the power 2: multiply the base by itself.
- *PEMDAS* gives the full work order. **P** means Parentheses. **E** means Exponents. **M** and **D** mean Multiply and Divide; they share one level, so work left to right. **A** and **S** mean Add and Subtract; they share one level, so work left to right.
- In this item, the parentheses make the full signed number -5 the base. Then the exponent tells us to square that whole base. That is why P and E control the first work.
- A *product* is the answer to a multiplication problem.
- Same signs multiply to a *positive* product (negative $\times$ negative = positive).
- When we say signs *cancel* here, we mean two negatives undo each other and leave a positive result.

Work the problem one tiny step at a time

1. Start with P in PEMDAS. The parentheses show the base is the full signed number -5.
2. Next use E in PEMDAS. The exponent 2 says to square that full base.
3. Write $(-5)^2 = (-5) \times (-5)$.
4. Look at the signs: both factors are negative. Same signs.

5. Negative times negative: the signs cancel to positive.
6. Multiply the sizes: $5 \times 5 = 25$.
7. So $(-5)^2 = +25$.
8. Check: $(-5) \times (-5) = 25$. The parentheses forced the base to stay negative before squaring.

Final answer: 25

Why this makes sense

Parentheses force you to square the negative number. Two negatives multiply to a positive: $(-5) \times (-5) = 25$. Compare carefully with Question 4, where missing parentheses change the order.

Question 4

Question

Evaluate: $-5^2 =$

Explain briefly how Question 3 and Question 4 differ.
Checked answer: -25 (parentheses change the order)

What the question is asking

This question asks for the value of -5^2 (minus, then five with a raised 2) and how it differs from $(-5)^2$ (curved marks wrap -5, then a raised 2).

Important words

- *PEMDAS* gives the work order: **P** means Parentheses; **E** means Exponents; **M** and **D** mean Multiply and Divide from left to right; **A** and **S** mean Add and Subtract from left to right.
- An *operation* is a math action. Add joins amounts. Subtract takes away or finds a difference. Multiply makes equal groups. Divide splits into equal groups.
- An *expression* is a math phrase made of numbers and operation signs.
- A *factor* is a number being multiplied.
- A *base* is the number used as a repeated factor.
- An *exponent* is the small raised number that tells how many times to use its base as a factor.
- To *square* a base means raise it to exponent 2, or multiply the base by itself. *Squared* means raised to exponent 2.
- The expression -5^2 means "the opposite of 5^2," not "negative five squared."
- The *base* here is just the digit 5 (not -5), because there are no parentheses wrapping the minus.
- To *negate* means take the opposite (put a minus in front) after the power is computed.
- *Parentheses* change which number gets squared. That is the whole difference from Question 3.
- Here there are no parentheses around -5. That is why PEMDAS makes us do the exponent before the front minus.

Work the problem one tiny step at a time

1. Look for parentheses around -5: there are none.
2. PEMDAS says to do E (Exponents) before applying the front minus.
3. The base that gets squared is 5, not -5.
4. Compute $5^2 = 5 \times 5 = 25$.
5. Then apply the minus (negate): $-5^2 = -25$.
6. Difference from Question 3: $(-5)^2$ squares the negative and gives $+25$. -5^2 squares 5 first, then makes it negative, giving -25.
7. Check side by side: $(-5) \times (-5) = 25$, but $-(5 \times 5) = -25$.

Final answer: -25

Why this makes sense

Parentheses change which number gets squared. Without them, square first, then negate. Check: $5^2 = 25$, then the front minus makes -25. A very common mistake is treating -5^2 like $(-5)^2$.

Lesson 1-7: Multiply Rational Numbers

Question 1

Question

Evaluate: $-0.5(10 + 6) =$
Checked answer: -8

What the question is asking

This question asks for the value of $-0.5(10 + 6)$. Do the add inside the curved marks first, then multiply.

Important words

- *Nonzero* means not equal to zero.
- A *fraction* $\frac{a}{b}$ means a equal parts out of b equal parts, where b is not zero.
- A *terminating decimal* stops. A *repeating decimal* goes forever in a repeating pattern.
- A *rational number* can be written as a fraction with a nonzero bottom number, or as a terminating or repeating decimal.
- *PEMDAS*: Parentheses first, then Exponents, then Multiply/Divide left to right, then Add/Subtract left to right.
- *Parentheses* ( ) group work that must be done first.
- A number written next to parentheses means *multiply*: $-0.5(10 + 6)$ means $-0.5 \times (10 + 6)$.
- A *decimal* 0.5 means one half, so 0.5×16 is half of 16.

- An *expression* is a math phrase to evaluate (find the value of).
- A *factor* is a number being multiplied. Here -0.5 and the parenthesized value are factors.
- Sign rule: negative times positive is *negative.*
- The leading minus on -0.5 is part of the factor being multiplied, not a separate subtract step after.

Work the problem one tiny step at a time

1. PEMDAS letter P: work inside parentheses first (before multiply).
2. Inside the parentheses: $10 + 6 = 16$.
3. Rewrite the expression: -0.5×16.
4. Multiply the sizes first (ignore the sign for a moment): 0.5 is one half.
5. Half of 16: $16 \div 2 = 8$, so $0.5 \times 16 = 8$.
6. Now apply the sign: -0.5 is negative and 16 is positive, so the product is negative.
7. Therefore $-0.5 \times 16 = -8$.
8. Check another way: $-0.5 \times 16 = -\frac{1}{2} \times 16 = -\frac{16}{2} = -8$. Same answer.

Final answer: -8

Why this makes sense

Parentheses force the add before the multiply. Half of 16 is 8, and the leading negative keeps the answer negative. Check: $(-0.5) \times 16 = -8$, and also $-\frac{16}{2} = -8$.

Lesson 1-8: Divide Integers

Question 1

Question

A submarine starts at sea level, 0 feet. It descends to -168 feet in 7 minutes at a steady rate.
Part A: What integer represents the unit rate of descent in feet per minute?
Part B: Where will the submarine be in relation to sea level after 4 minutes at this same rate?
Checked answer: Part A: -24; Part B: -96 feet (below sea level)

What the question is asking

This question asks for the up-or-down change for one minute of going down (down is negative), then the position after 4 minutes at that same per-minute change.

Important words

- *Sea level* is 0 feet. Below sea level is negative.
- *Descending* means going down, so the change is negative.
- A *unit* is one single amount. Here the time unit is one minute.

- A *unit rate* is the change for one unit of time (one minute).
- A *decimal* uses a decimal point to show parts of a whole.
- An *integer* is one of the numbers $\ldots, -2, -1, 0, 1, 2, \ldots$; it has no decimal part.
- *Divide* change by time to find the rate: rate $= \dfrac{\text{change}}{\text{time}}$.
- In a division $a \div b$, the *dividend* is the number being divided (a), and the *divisor* is the number you divide by (b).
- A *quotient* is the answer to a division problem.
- Sign rule for division: opposite signs make a negative quotient.
- Sign rule for multiplication: negative times positive is negative.

Work the problem one tiny step at a time

1. **Part A:** Total change is -168 feet in 7 minutes.
2. Unit rate: $(-168) \div 7$.
3. Name the parts: dividend -168, divisor 7.
4. Ignore signs: $168 \div 7 = 24$ (because $7 \times 24 = 168$).
5. Opposite signs: answer is negative. Rate $= -24$ feet per minute.
6. Check Part A: $(-24) \times 7 = -168$. Matches the given descent.
7. **Part B:** Multiply rate by 4 minutes: $(-24) \times 4$.
8. Size: $24 \times 4 = 96$, then keep the negative: -96.
9. So the submarine is at -96 feet (96 feet below sea level).
10. Check Part B: four minutes of -24 each is $(-24) + (-24) + (-24) + (-24) = -96$.

Final answer: Part A: -24; Part B: -96 feet (below sea level)

Why this makes sense
Steady rate means the same drop each minute. Four minutes at -24 per minute is -96. Check: $(-24) \times 7 = -168$ recovers the original descent, and $(-24) \times 4 = -96$ is the 4-minute position.

Question 2

Question
Evaluate: $(-96) \div 8 =$
Checked answer: -12

What the question is asking
This question asks for the division answer of $(-96) \div 8$.

Important words

- An *operation* is a math action. Add joins amounts. Subtract takes away or finds a difference. Multiply makes equal groups. Divide splits into equal groups.
- An *expression* is a math phrase made of numbers and operation signs.
- A *quotient* is the answer to a division problem.
- *Division* splits a total into equal groups: here, how many groups of 8 fit into 96, then apply the sign.
- In a division $a \div b$, the *dividend* is the number being divided (a), and the *divisor* is the number you divide by (b). Here the dividend is -96 and the divisor is 8.
- To *evaluate* means find the value of the expression.
- A *product* is the answer to a multiplication problem.
- *Division sign rule*:
  - Opposite signs (one negative, one positive) make a **negative** quotient.
  - Same signs (both positive or both negative) make a **positive** quotient.
- A good *check*: multiply the quotient by the divisor; you should get back the dividend.

Work the problem one tiny step at a time

1. Write the problem: $(-96) \div 8$.
2. Name the parts: dividend -96, divisor 8.
3. Look at the signs: -96 is negative and 8 is positive. These are **opposite** signs.
4. Opposite signs mean the quotient will be negative (after we find the size).
5. Ignore signs and divide the sizes: $96 \div 8$.
6. Think: $8 \times 12 = 96$, so $96 \div 8 = 12$.
7. Put the negative sign on the quotient: $(-96) \div 8 = -12$.
8. Check by multiplying: $(-12) \times 8$.
9. Size: $12 \times 8 = 96$. Opposite signs: product is negative. So $(-12) \times 8 = -96$. Check works: quotient $\times$ divisor $=$ dividend.

Final answer: -12

Why this makes sense

Opposite signs in division give a negative answer. The size is $96 \div 8 = 12$, so the signed quotient is -12. Multiplying back $(-12) \times 8 = -96$ confirms the work. A common mistake is dropping the negative and writing only 12.

Question 3

Question

Evaluate: $(-72) \div (-9) =$
Checked answer: 8

What the question is asking

This question asks for the division answer of $(-72) \div (-9)$.

Important words

- An *operation* is a math action. Add joins amounts. Subtract takes away or finds a difference. Multiply makes equal groups. Divide splits into equal groups.
- An *expression* is a math phrase made of numbers and operation signs.
- A *quotient* is the answer to a division problem.
- *Division* means splitting into equal groups or finding how many equal groups fit.
- In a division $a \div b$, the *dividend* is the number being divided (a), and the *divisor* is the number you divide by (b). Here the dividend is -72 and the divisor is -9.
- To *evaluate* means find the value of the expression.
- A *product* is the answer to a multiplication problem.
- *Division sign rule*:
  - Same signs (both negative here) make a **positive** quotient.
  - Opposite signs make a **negative** quotient.
- Two negatives in division work like two negatives in multiplication: the signs cancel to positive.
- To *cancel signs* (here) means two negatives undo each other and leave a positive result.
- A good *check*: multiply the quotient by the divisor; you should get back the dividend.

Work the problem one tiny step at a time

1. Write the problem: $(-72) \div (-9)$.
2. Name the parts: dividend -72, divisor -9.
3. Look at the signs: both numbers are negative. These are the **same** signs.
4. Same signs mean the quotient will be positive (after we find the size).
5. Ignore signs and divide the sizes: $72 \div 9$.
6. Think: $9 \times 8 = 72$, so $72 \div 9 = 8$.
7. Put the positive sign on the quotient: $(-72) \div (-9) = 8$.
8. Check by multiplying: $8 \times (-9)$.

9. Size: $8 \times 9 = 72$. Opposite signs: product is negative. So $8 \times (-9) = -72$. Check works: quotient $\times$ divisor $=$ dividend.

Final answer: 8

Why this makes sense

Two negatives in division cancel to a positive, just like two negatives in multiplication. The size is $72 \div 9 = 8$, so the signed quotient is $+8$. Check: $8 \times (-9) = -72$. A common mistake is thinking two negatives in division still make a negative.

Question 4

Question

Find the value of n: $\dfrac{-18}{6} = \dfrac{n}{-4}$

Checked answer: $n = 12$

What the question is asking

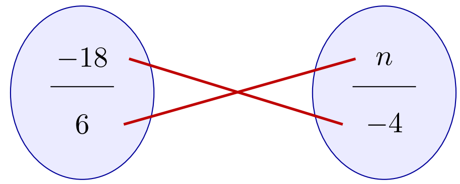
This question asks for the number n that makes the two fractions equal.

Important words

- A *variable* n is a letter that stands for an unknown number.
- *Nonzero* means not equal to zero.
- A *fraction* is a top number divided by a nonzero bottom number. The fraction bar means divide.
- Two fractions are *equal* when they represent the same value.
- A *diagonal* is a slanted line joining opposite entries in the two fractions.
- The *butterfly method* (cross multiply) multiplies along each diagonal of equal fractions $\dfrac{a}{b} = \dfrac{c}{d}$; you get $a \cdot d = b \cdot c$.
- *Cross products* are the two diagonal multiplies from the butterfly.
- An *operation* is a math action. Add joins amounts. Subtract takes away or finds a difference. Multiply makes equal groups. Divide splits into equal groups.
- An *expression* is a math phrase made of numbers and operation signs.
- An *equation* says two sides are equal. You may do the same operation to both sides.
- To *simplify* means rewrite a math expression in a shorter form with the same value.

Work the problem one tiny step at a time

1. Write the equal fractions: $\dfrac{-18}{6} = \dfrac{n}{-4}$.
2. Draw the butterfly. The red lines are the diagonals.



Butterfly: multiply along each red line.

3. Multiply along one red line: $(-18) \times (-4)$.
4. Negative times negative is positive: $(-18) \times (-4) = 72$.
5. Multiply along the other red line: $6 \times n = 6n$.
6. Set the cross products equal: $6n = 72$.
7. Divide both sides by 6 (same operation to both sides of the equals sign):

$$6n = 72$$
$$\underline{\div 6 \qquad \div 6}$$

8. Simplify the left side: $6n \div 6 = n$. Dividing by 6 undoes the multiply-by-6.
9. Now divide the right side: $72 \div 6$.
10. Break 72 into $60 + 12$.
11. Divide the first part: $60 \div 6 = 10$.
12. Divide the second part: $12 \div 6 = 2$.
13. Add the two quotients: $10 + 2 = 12$.
14. So $n = 12$.
15. Check the left fraction: $(-18) \div 6 = -3$.
16. Check the right fraction: $12 \div (-4) = -3$.
17. Both fractions equal -3, so the check works.

Final answer: $n = 12$

Why this makes sense

Equal fractions have equal cross products. Both sides simplify to -3, so $n = 12$ is correct. A common mistake is mixing up which numbers multiply.

Lesson 1-9: Divide Rational Numbers

Question 1

Question

A business loses \$450.75 as a result of a shipping delay. The 3 owners of the business have to share the loss equally. Write the quotient that represents each person's share of the loss.
Checked answer: -150.25 (or $-\$150.25$ per owner)

What the question is asking

This question asks for each owner's equal share of a \$450.75 loss as a division answer that keeps the loss sign (negative).

Important words

- A *decimal* uses a decimal point to show parts of a whole.
- The *decimal point* is the dot that separates whole-number places from parts of a whole.
- A *loss* is money going out, so we write it as a *negative* number.
- To share *equally* means *divide* by the number of people.
- A *quotient* is the answer to a division problem.
- *Long division* is a written way to divide one digit at a time.
- A *remainder* is the amount left after one division step.
- To *bring down* a digit means move the next digit beside the remainder so division can continue.
- In *decimal division*, place the decimal point in the answer directly above the decimal point in the number being divided.
- Division sign rule: opposite signs make a negative quotient; same signs make a positive quotient.

Work the problem one tiny step at a time

1. Write the loss as -450.75.
2. There are 3 owners, so divide: $(-450.75) \div 3$.
3. Ignore the sign first and divide $450.75 \div 3$.
4. 3 goes into 4 one time (3), remainder 1.
5. Bring down 5 to make 15. $3 \times 5 = 15$. Remainder 0.
6. Bring down the 0 from 450. 3 goes into 0 zero times, so write 0 as the next quotient digit.
7. The next mark in 450.75 is the decimal point. Put a decimal point in the quotient directly above it.
8. Bring down 7. 3 goes into 7 two times (6), remainder 1.

9. Bring down 5 to make 15. $3 \times 5 = 15$. Remainder 0.
10. So $450.75 \div 3 = 150.25$.
11. Opposite signs: the quotient is negative. Each share is -150.25.

Final answer: -150.25 (or $-\$150.25$ per owner)

Why this makes sense

Three equal shares of a loss stay negative. Check: $(-150.25) \times 3 = -450.75$.

Question 2

Question

A race is $15\frac{1}{2}$ miles long. The organizers want to put a water station every $3\frac{7}{8}$ miles along the route, and one at the finish line. How many total water stations are needed?
Checked answer: 4 water stations

What the question is asking

This question asks how many water stations are needed if one is placed every $3\frac{7}{8}$ miles on a $15\frac{1}{2}$-mile race, including the finish.

Important words

- *Nonzero* means not equal to zero.
- An *operation* is a math action. Add joins amounts. Subtract takes away or finds a difference. Multiply makes equal groups. Divide splits into equal groups.
- An *expression* is a math phrase made of numbers and operation signs.
- A *whole number* has no fractional part, such as 15.
- A *fraction* has a top number and a nonzero bottom number and names equal parts of a whole.
- The *numerator* is the top number of a fraction.
- The *denominator* is the bottom number of a fraction; it tells how many equal parts make one whole.
- A *factor* is a number being multiplied.
- A *mixed number* has a whole number and a fraction, like $15\frac{1}{2}$.
- An *improper fraction* has a numerator larger than or equal to the denominator, like $\frac{31}{2}$.
- To change a mixed number $a\frac{b}{c}$ to improper: $\dfrac{a \cdot c + b}{c}$.
- The *reciprocal* of a fraction flips the numerator and the denominator (swap top$\leftrightarrow$bottom). Example: the reciprocal of $\frac{31}{8}$ is $\frac{8}{31}$.
- To *flip* a fraction means swap numerator and denominator.
- To *divide fractions*, multiply by the reciprocal: $\dfrac{a}{b} \div \dfrac{c}{d} = \dfrac{a}{b} \times \dfrac{d}{c}$.

- To *cancel matching factors* means remove the same factor from a numerator and a denominator before multiplying. Example: a 31 on top and a 31 on bottom cancel to 1.
- An *interval* is one spacing gap between stations.
- To *simplify* means rewrite a math expression in a shorter form with the same value.

Work the problem one tiny step at a time

1. Convert race length: $15\frac{1}{2} = \dfrac{15 \times 2 + 1}{2} = \dfrac{31}{2}$.
2. Convert spacing: $3\frac{7}{8} = \dfrac{3 \times 8 + 7}{8} = \dfrac{31}{8}$.
3. Divide distance by spacing to count intervals: $\dfrac{31}{2} \div \dfrac{31}{8}$.
4. Write the reciprocal of the second fraction: flip $\dfrac{31}{8}$ (swap top and bottom) to get $\dfrac{8}{31}$.
5. Multiply by that reciprocal: $\dfrac{31}{2} \times \dfrac{8}{31}$.
6. Cancel the matching 31s (same factor top and bottom): $\dfrac{31}{2} \times \dfrac{8}{31}$ becomes $\dfrac{1}{2} \times \dfrac{8}{1}$.
7. Multiply: $\dfrac{1}{2} \times \dfrac{8}{1} = \dfrac{8}{2}$.
8. Simplify: $\dfrac{8}{2} = 4$.
9. That means 4 equal intervals along the race.
10. Stations are at the end of each interval, including the finish line. So there are 4 stations.

Final answer: 4 water stations

Why this makes sense

Four intervals of $3\frac{7}{8}$ fit exactly into $15\frac{1}{2}$. One station at the end of each interval gives 4 stations.

Lesson 1-10: Solve Problems with Rational Numbers

Question 1

Question

Two people are hiking. The first person climbs to the top of a peak that is 450 feet above sea level. The second person descends into a valley that is 120 feet below sea level. What is the total vertical distance between the two hikers?
Checked answer: 570 feet

What the question is asking

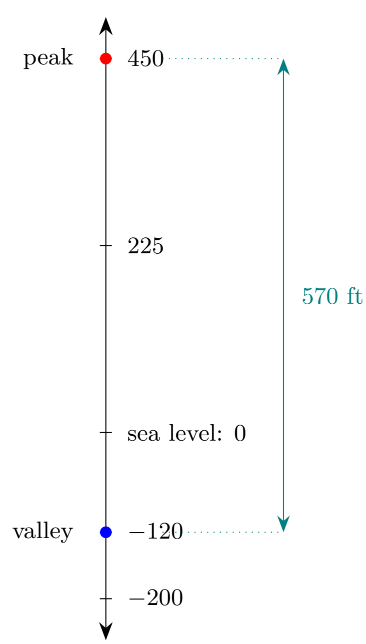
This question asks how many feet separate the two hikers when one is above zero and the other is below zero.

Important words

- A *negative number* is less than zero. A *positive number* is greater than zero.
- A *number line* is a straight line with numbers in order. Negative numbers are below or left of zero; positive numbers are above or right of zero.
- *Above sea level* is positive. *Below sea level* is negative.
- *Vertical* means up and down (not side to side).
- *Vertical distance* is how far apart two heights are on an up-down number line.
- *Sea level* is 0 feet (the zero mark between above and below).
- *Absolute value* means distance from zero on a number line. Distance is never negative.
- A *difference* is the answer to subtraction. To *subtract* means take away or find how far apart two numbers are.
- A distance can be found with $|a-b|$: subtract the two positions, then take absolute value. Subtracting a negative adds: $a-(-b)=a+b$.

Work the problem one tiny step at a time

1. Write the positions: $+450$ (peak) and -120 (valley).
2. Distance $= |450-(-120)|$.
3. Rewrite the subtraction: $450-(-120)=450+120$.



4. Add the ones: $0+0=0$.
5. Add the tens: $5+2=7$, so $50+20=70$.

6. Add the hundreds: $400 + 100 = 500$.
7. Total: $500 + 70 = 570$.
8. Absolute value of 570 is still 570.
9. Think: 450 feet up from sea level plus 120 feet down from sea level is 570 feet apart.

Final answer: 570 feet

Why this makes sense

One hiker is above zero and one is below zero, so you add the two distances from sea level: $450 + 120 = 570$. Check: $|450 - (-120)| = |570| = 570$.

Lesson SC-2: Use the Laws of Exponents

Question 1

Question

A person is designing a rectangular poster. The area of the poster is given as the expression $3^2 \times 3^4$ square inches. Use the laws of exponents to simplify the expression and find the total area of the poster.
Checked answer: 729 square inches

What the question is asking

This question asks you to find the number of square inches shown by $3^2 \times 3^4$.

Important words

- An *expression* is a math phrase made of numbers and operation signs.
- To *simplify* means rewrite an expression in a shorter form with the same value.
- A *factor* is a number being multiplied.
- The *base* is the number used as a repeated factor. In 3^2, the base is 3.
- An *exponent* is the small raised number that tells how many times to use the base as a factor.
- A *power* is a base and its exponent together, such as 3^2.
- A *product* is the answer to a multiplication problem.
- The *product-of-powers law* says that when powers have the same base, keep the base and add the exponents: $a^m \times a^n = a^{m+n}$.
- To *expand* a power means write all of its repeated factors.
- *Area* means the amount of flat space covered. It is measured here in square inches.

Work the problem one tiny step at a time

1. Look at the first power, 3^2. Its base is 3.
2. Look at the second power, 3^4. Its base is also 3.
3. The bases match, so the product-of-powers law applies.
4. Keep the base 3.
5. Add the exponents: $2 + 4 = 6$.
6. Rewrite the expression: $3^2 \times 3^4 = 3^6$.
7. Expand the power: $3^6 = 3 \times 3 \times 3 \times 3 \times 3 \times 3$.
8. Multiply the first two factors: $3 \times 3 = 9$.
9. Multiply by the next 3: $9 \times 3 = 27$.
10. Multiply by the next 3: $27 \times 3 = 81$.
11. Multiply by the next 3: $81 \times 3 = 243$.
12. Multiply by the last 3: $243 \times 3 = 729$.

Final answer: 729 square inches

Why this makes sense

The two powers contain two factors of 3 and four more factors of 3. That makes six factors of 3 altogether. Expanding 3^6 gives 729, so the poster area is 729 square inches.

Question 2

Question

A person is baking cupcakes and makes a total of 2^4 cupcakes. They want to divide the batch equally among 2^2 friends.
Part A: What expression gives the number of cupcakes each friend will get?
Part B: How many cupcakes will each friend get?

Checked answer: Part A: $\dfrac{2^4}{2^2}$ (or 2^{4-2} or 2^2); Part B: 4

What the question is asking

This question asks you to write the sharing problem and find how many cupcakes one friend receives.

Important words

- An *operation* is a math action. Add joins amounts. Subtract takes away or finds a difference. Multiply makes equal groups. Divide splits into equal groups.
- An *expression* is a math phrase made of numbers and operation signs.
- *Equivalent* expressions look different but have the same value.
- To share *equally* means every friend gets the same amount.

- To *divide* means split a total into equal groups.
- The *dividend* is the total being divided. Here it is 2^4 cupcakes.
- The *divisor* is the number of equal groups. Here it is 2^2 friends.
- A *quotient* is the answer to a division problem.
- A *factor* is a number being multiplied.
- The *base* is the number used as a repeated factor. The base is 2 in both powers.
- An *exponent* is the small raised number that tells how many times to use the base as a factor.
- A *power* is a base and its exponent together, such as 2^4.
- *Nonzero* means not equal to zero.
- The *quotient-of-powers law* says that when powers have the same nonzero base, keep the base and subtract the bottom exponent from the top exponent: $\dfrac{a^m}{a^n} = a^{m-n}$.

Work the problem one tiny step at a time

1. The total number of cupcakes is 2^4.
2. The number of friends is 2^2.
3. Sharing equally means divide the total by the number of friends.
4. Write Part A: $\dfrac{2^4}{2^2}$.
5. Both powers have base 2, so use the quotient-of-powers law.
6. Subtract the exponents: $4 - 2 = 2$.
7. Rewrite the quotient: $\dfrac{2^4}{2^2} = 2^2$.
8. Expand 2^2: $2^2 = 2 \times 2$.
9. Multiply: $2 \times 2 = 4$.
10. Check by finding both powers: $2^4 = 2 \times 2 \times 2 \times 2 = 16$ and $2^2 = 4$.
11. Divide to check: $16 \div 4 = 4$.

Final answer: Part A: $\dfrac{2^4}{2^2}$ (or an equivalent expression); Part B: 4 cupcakes per friend

Why this makes sense

Sixteen cupcakes split among four friends gives four cupcakes to each friend. The exponent law and ordinary division agree.

Question 3

Question
Evaluate using the power-of-a-power law: $(4^3)^2 =$ _________.
Checked answer: 4096

What the question is asking

This question asks you to find the number represented by $(4^3)^2$.

Important words

- To *evaluate* means find the number value.
- *Parentheses* are curved marks that group math together.
- A *factor* is a number being multiplied.
- The *base* is the number used as a repeated factor. Here the base is 4.
- An *exponent* is the small raised number that tells how many times to use the base as a factor.
- A *power* is a base and its exponent together, such as 4^3.
- A *power of a power* has one exponent on a power and another exponent outside the parentheses.
- The *power-of-a-power law* says to keep the base and multiply the exponents: $(a^m)^n = a^{m\times n}$.
- To *expand* a power means write all of its repeated factors.

Work the problem one tiny step at a time

1. In $(4^3)^2$, the base is 4.
2. The exponent inside the parentheses is 3.
3. The exponent outside the parentheses is 2.
4. Use the power-of-a-power law.
5. Multiply the exponents: $3 \times 2 = 6$.
6. Keep the base 4: $(4^3)^2 = 4^6$.
7. Expand: $4^6 = 4 \times 4 \times 4 \times 4 \times 4 \times 4$.
8. Multiply the first two factors: $4 \times 4 = 16$.
9. Multiply by the next 4: $16 \times 4 = 64$.
10. Multiply by the next 4: $64 \times 4 = 256$.
11. Multiply by the next 4: $256 \times 4 = 1024$.
12. Multiply by the last 4: $1024 \times 4 = 4096$.
13. Check another way: $4^3 = 64$, and $64^2 = 64 \times 64 = 4096$.

Final answer: 4096

Why this makes sense

The outside exponent 2 means use the whole power 4^3 twice. That makes six factors of 4, so multiplying the exponents gives the same value as expanding.

Question 4

Question

Evaluate: $7^0 =$ __________.
Checked answer: 1

What the question is asking

This question asks for the number represented by 7 with a raised 0.

Important words

- To *evaluate* means find the number value.
- The *base* is the number used as a repeated factor. Here the base is 7.
- An *exponent* is the small raised number that tells how many times to use the base as a factor. Here the exponent is 0.
- A *power* is a base and its exponent together, such as 7^0.
- *Nonzero* means not equal to zero.
- The *zero-exponent law* says any nonzero base raised to the power 0 equals 1: $a^0 = 1$ when $a \neq 0$.

Work the problem one tiny step at a time

1. Check the base: the base is 7.
2. Seven is not zero, so the zero-exponent law applies.
3. Follow the power pattern to see why.
4. Start with $7^2 = 7 \times 7 = 49$.
5. Lower the exponent by 1 and divide by 7: $49 \div 7 = 7$, so $7^1 = 7$.
6. Lower the exponent by 1 again and divide by 7: $7 \div 7 = 1$, so $7^0 = 1$.
7. Apply the law directly: $7^0 = 1$.

Final answer: 1

Why this makes sense

The power pattern divides by 7 each time the exponent drops by 1. The step from 7^1 to 7^0 is $7 \div 7 = 1$. A zero exponent does not make the answer zero.

Question 5

Question

Simplify $\frac{4^5}{4^2} \cdot 4^0 \cdot 4^{-1}$.

Checked answer: 16

What the question is asking

This question asks you to find the number represented by the powers of 4 in the expression.

Important words

- An *operation* is a math action. Add joins amounts. Subtract takes away or finds a difference. Multiply makes equal groups. Divide splits into equal groups.
- An *expression* is a math phrase made of numbers and operation signs.
- To *simplify* means rewrite an expression in a shorter form with the same value.
- To *evaluate* means find the number value of an expression.
- A *factor* is a number being multiplied.
- The *base* is the number used as a repeated factor. Every power here has base 4.
- An *exponent* is the small raised number that tells how many times to use the base as a factor.
- A *power* is a base and its exponent together, such as 4^5.
- A *quotient* is the answer to a division problem.
- *Nonzero* means not equal to zero.
- The *quotient-of-powers law* says to keep a matching nonzero base and subtract exponents: $\frac{a^m}{a^n} = a^{m-n}$.
- The *zero-exponent law* says $a^0 = 1$ for every nonzero base.
- A *fraction* $\frac{a}{b}$ means a equal parts out of b equal parts, where b is not zero.
- A *reciprocal* is made by flipping a nonzero fraction. The reciprocal of $\frac{4}{1}$ is $\frac{1}{4}$.
- A *positive exponent* is an exponent greater than zero. A *positive power* uses a positive exponent, such as 4^1.
- A *negative exponent* tells you to write the reciprocal of the matching positive power: $a^{-n} = \frac{1}{a^n}$.
- A *product* is the answer to multiplication.
- *PEMDAS* gives the work order: Parentheses, Exponents, Multiply and Divide from left to right, then Add and Subtract from left to right.

Work the problem one tiny step at a time

1. PEMDAS has no parentheses to work here.
2. Work with the powers before completing the multiplication and division.
3. Start with the quotient $\dfrac{4^5}{4^2}$.
4. The bases match, so subtract the exponents: $5 - 2 = 3$.
5. Rewrite the quotient: $\dfrac{4^5}{4^2} = 4^3$.
6. Use the zero-exponent law: $4^0 = 1$.
7. Rewrite the negative exponent: $4^{-1} = \dfrac{1}{4^1}$.
8. Evaluate $4^1 = 4$, so $4^{-1} = \dfrac{1}{4}$.
9. The expression is now $4^3 \cdot 1 \cdot \dfrac{1}{4}$.
10. Expand 4^3: $4 \times 4 \times 4 = 64$.
11. Multiply by 1: $64 \cdot 1 = 64$.
12. Multiply by one fourth: $64 \cdot \dfrac{1}{4} = \dfrac{64}{4}$.
13. Divide: $64 \div 4 = 16$.

Final answer: 16

Why this makes sense

The quotient leaves three factors of 4. The negative exponent contributes a factor of one fourth, which removes one of those factors. Two factors of 4 remain, and $4^2 = 16$.

Question 6

Question

Simplify $(2^3)^2 \cdot (3 \cdot 2)^2 \div \left(\dfrac{12}{2}\right)^2$.

Checked answer: 64

What the question is asking

This question asks you to find the number represented by the whole expression.

Important words

- An *operation* is a math action. Add joins amounts. Subtract takes away or finds a difference. Multiply makes equal groups. Divide splits into equal groups.
- An *expression* is a math phrase made of numbers and operation signs.
- To *simplify* means rewrite an expression in a shorter form with the same value.

- To *evaluate* means find the number value of an expression.
- *Parentheses* are curved marks that group work together.
- A *factor* is a number being multiplied.
- The *base* is the number used as a repeated factor.
- An *exponent* is the small raised number that tells how many times to use the base as a factor.
- A *power* is a base and its exponent together, such as 2^3.
- To *expand* a power means write all of its repeated factors.
- *Nonzero* means not equal to zero.
- To *multiply* means combine equal groups. A *product* is the answer to multiplication.
- The *power-of-a-power law* says $(a^m)^n = a^{m\times n}$: keep the base and multiply the two exponents.
- The *power-of-a-product law* says $(ab)^n = a^n b^n$: raise each factor in the product to the exponent.
- To *divide* means split into equal groups. A *quotient* is the answer to division. In a/b, a is divided by b.
- The *power-of-a-quotient law* says $\left(\dfrac{a}{b}\right)^n = \dfrac{a^n}{b^n}$ when $b \neq 0$.
- The condition $b \neq 0$ says b must be nonzero.
- *Equal rank* means two operations share one place in the work order. Multiply and divide have equal rank, so do them from left to right.
- To *cancel* matching nonzero factors means divide them to make 1.
- *PEMDAS* gives the order: Parentheses, Exponents, Multiply and Divide from left to right, then Add and Subtract from left to right.

Work the problem one tiny step at a time

1. PEMDAS letter P says work inside parentheses first when possible.
2. Inside the second group, multiply: $3 \cdot 2 = 6$.
3. Inside the third group, divide: $12 \div 2 = 6$.
4. Rewrite the expression: $(2^3)^2 \cdot 6^2 \div 6^2$.
5. PEMDAS letter E says work with exponents next.
6. Use power of a power: $(2^3)^2 = 2^{3\times 2}$.
7. Multiply the exponents: $3 \times 2 = 6$.
8. Rewrite the power: $(2^3)^2 = 2^6$.
9. Expand: $2^6 = 2 \times 2 \times 2 \times 2 \times 2 \times 2$.

10. Multiply the first two factors: $2 \times 2 = 4$.
11. Multiply by the next 2: $4 \times 2 = 8$.
12. Multiply by the next 2: $8 \times 2 = 16$.
13. Multiply by the next 2: $16 \times 2 = 32$.
14. Multiply by the last 2: $32 \times 2 = 64$.
15. Expand the other power: $6^2 = 6 \times 6$.
16. Multiply: $6 \times 6 = 36$.
17. The expression is now $64 \cdot 36 \div 36$.
18. PEMDAS says multiplication and division have equal rank, so work from left to right.
19. To multiply $64 \cdot 36$, break 36 into $30 + 6$.
20. Start $64 \cdot 30$ by finding $64 \cdot 3$.
21. Multiply the tens: $60 \cdot 3 = 180$.
22. Multiply the ones: $4 \cdot 3 = 12$.
23. Add: $180 + 12 = 192$.
24. Multiply by 10: $192 \cdot 10 = 1920$.
25. So $64 \cdot 30 = 1920$.
26. Now find $64 \cdot 6$.
27. Multiply the tens: $60 \cdot 6 = 360$.
28. Multiply the ones: $4 \cdot 6 = 24$.
29. Add: $360 + 24 = 384$.
30. Add the two partial products: $1920 + 384 = 2304$.
31. Now divide $2304 \div 36$. Break 2304 into $2160 + 144$.
32. Multiply: $36 \cdot 6 = 216$.
33. Multiply by 10: $36 \cdot 60 = 2160$.
34. So $2160 \div 36 = 60$.
35. Multiply: $36 \cdot 4 = 144$.
36. So $144 \div 36 = 4$.
37. Add the two quotients: $60 + 4 = 64$.
38. Check the matching factors: $36 \div 36 = 1$.

39. Multiply: $64 \cdot 1 = 64$.

Final answer: 64

Why this makes sense

The second power and the third power are both $6^2 = 36$. Multiplying by 36 and then dividing by 36 leaves the first value unchanged. The first value is $(2^3)^2 = 64$.

Question 7

Question

Simplify $5^0 + 3^{-2} + \dfrac{2^6 \cdot 2^3}{2^7}$.

Checked answer: $5\frac{1}{9}$

What the question is asking

This question asks you to find the number made by adding the three parts of the expression.

Important words

- An *operation* is a math action. Add joins amounts. Subtract takes away or finds a difference. Multiply makes equal groups. Divide splits into equal groups.
- An *expression* is a math phrase made of numbers and operation signs.
- To *simplify* means rewrite an expression in a shorter form with the same value.
- To *evaluate* means find the number value of an expression.
- A *factor* is a number being multiplied.
- *Nonzero* means not equal to zero.
- A *fraction* has a top number and a nonzero bottom number and names equal parts of a whole.
- A *whole number* has no fractional part, such as 5.
- The *numerator* is the top number of a fraction. The *denominator* is its bottom number.
- The *base* is the number used as a repeated factor.
- An *exponent* is the small raised number that tells how many times to use the base as a factor.
- A *power* is a base and its exponent together, such as 5^0.
- The *zero-exponent law* says $a^0 = 1$ when $a \neq 0$.
- To *flip* a fraction means swap its numerator (top number) and denominator (bottom number).
- A *reciprocal* is the result of flipping a nonzero fraction. The reciprocal multiplies with the original fraction to make 1. For example, flip $\dfrac{9}{1}$ to get its reciprocal $\dfrac{1}{9}$.
- A *negative exponent* tells you to use the reciprocal: $a^{-n} = \dfrac{1}{a^n}$.
- To *multiply* means combine equal groups. A *product* is the answer to multiplication.

- The *product-of-powers law* says $a^m \cdot a^n = a^{m+n}$ for matching bases.
- To *divide* means split into equal groups. A *quotient* is the answer to division.
- The *quotient-of-powers law* says $\dfrac{a^m}{a^n} = a^{m-n}$ for a matching nonzero base.
- A *proper fraction* has a top number smaller than its bottom number, such as $\dfrac{1}{9}$.
- A *mixed number* has a whole number and a proper fraction, such as $5\frac{1}{9}$.
- *PEMDAS* means Parentheses, Exponents, Multiply and Divide from left to right, then Add and Subtract from left to right.

Work the problem one tiny step at a time

1. PEMDAS says to work with powers before the final addition.
2. Use the zero-exponent law: $5^0 = 1$.
3. Rewrite the negative exponent: $3^{-2} = \dfrac{1}{3^2}$.
4. Evaluate the positive power: $3^2 = 3 \times 3 = 9$.
5. Therefore $3^{-2} = \dfrac{1}{9}$.
6. In the numerator of the fraction, multiply powers with base 2.
7. Add the numerator exponents: $6 + 3 = 9$.
8. Rewrite the numerator: $2^6 \cdot 2^3 = 2^9$.
9. Divide powers with base 2: $\dfrac{2^9}{2^7} = 2^{9-7}$.
10. Subtract the exponents: $9 - 7 = 2$.
11. Evaluate: $2^2 = 2 \times 2 = 4$.
12. The full expression is now $1 + \dfrac{1}{9} + 4$.
13. Add the whole numbers: $1 + 4 = 5$.
14. Keep the fractional part: $5 + \dfrac{1}{9} = 5\frac{1}{9}$.

Final answer: $5\frac{1}{9}$

Why this makes sense

The three parts are 1, one ninth, and 4. The whole-number parts total 5, and the small positive fraction adds one ninth more. That gives $5\frac{1}{9}$.

Question 8

Question

Simplify $\frac{7^4}{7^2} \cdot 7^0$.

Checked answer: 49

What the question is asking

This question asks you to find the number represented by the powers of 7.

Important words

- *PEMDAS* gives the work order: **P** means Parentheses; **E** means Exponents; **M** and **D** mean Multiply and Divide from left to right; **A** and **S** mean Add and Subtract from left to right.
- An *operation* is a math action. Add joins amounts. Subtract takes away or finds a difference. Multiply makes equal groups. Divide splits into equal groups.
- An *expression* is a math phrase made of numbers and operation signs.
- To *simplify* means rewrite an expression in a shorter form with the same value.
- A *factor* is a number being multiplied.
- The *base* is the number used as a repeated factor. Each power here has base 7.
- An *exponent* is the small raised number that tells how many times to use the base as a factor.
- A *power* is a base and its exponent together, such as 7^4.
- *Nonzero* means not equal to zero.
- The *quotient-of-powers law* says to keep a matching nonzero base and subtract exponents: $\frac{a^m}{a^n} = a^{m-n}$.
- The *zero-exponent law* says every nonzero base to the power 0 equals 1.
- A *product* is the answer to multiplication.
- A *quotient* is the answer to division.
- To *expand* a power means write all of its repeated factors.
- To *cancel matching factors* means divide the same nonzero factor from the top and bottom, which leaves a factor of 1.

Work the problem one tiny step at a time

1. PEMDAS says E (Exponents) comes before multiplication or division. Evaluate every power first.
2. Expand 7^4: $7 \times 7 \times 7 \times 7$.
3. Multiply the first pair: $7 \times 7 = 49$.
4. Multiply the second pair: $7 \times 7 = 49$.

5. Multiply the two results: $49 \times 49 = 2401$.
6. Expand 7^2: 7×7.
7. Multiply: $7 \times 7 = 49$.
8. Use the zero-exponent law: $7^0 = 1$.
9. Rewrite the expression after all exponent work: $2401 \div 49 \cdot 1$.
10. PEMDAS gives M and D equal rank, so work from left to right. Division appears first.
11. Divide: $2401 \div 49 = 49$ because $49 \times 49 = 2401$.
12. Multiply: $49 \cdot 1 = 49$.
13. Check with the quotient law: $7^4 \div 7^2 = 7^{4-2} = 7^2 = 49$.

Final answer: 49

Why this makes sense

Dividing four factors of 7 by two factors of 7 leaves two factors of 7. The zero power contributes 1, so it does not change the product.

Question 9

Question

Simplify $(3^2)^3$.
Checked answer: 729

What the question is asking

This question asks you to find the number represented by a power raised to another power.

Important words

- An *operation* is a math action. Add joins amounts. Subtract takes away or finds a difference. Multiply makes equal groups. Divide splits into equal groups.
- An *expression* is a math phrase made of numbers and operation signs.
- To *simplify* means rewrite an expression in a shorter form with the same value.
- A *factor* is a number being multiplied.
- The *base* is the number used as a repeated factor. Here the base is 3.
- An *exponent* is the small raised number that tells how many times to use the base as a factor.
- A *power* is a base and its exponent together, such as 3^2.
- A *power of a power* has an exponent outside parentheses that already contain a power.
- The *power-of-a-power law* says to keep the base and multiply the exponents: $(a^m)^n = a^{m \times n}$.
- To *expand* means write all repeated factors.

Work the problem one tiny step at a time

1. Find the inner exponent: it is 2.
2. Find the outer exponent: it is 3.
3. Use the power-of-a-power law.
4. Multiply the exponents: $2 \times 3 = 6$.
5. Keep the base 3: $(3^2)^3 = 3^6$.
6. Expand: $3^6 = 3 \times 3 \times 3 \times 3 \times 3 \times 3$.
7. Multiply: $3 \times 3 = 9$.
8. Multiply: $9 \times 3 = 27$.
9. Multiply: $27 \times 3 = 81$.
10. Multiply: $81 \times 3 = 243$.
11. Multiply: $243 \times 3 = 729$.
12. Check: $3^2 = 9$, so $(3^2)^3 = 9^3 = 9 \times 9 \times 9 = 81 \times 9 = 729$.

Final answer: 729

Why this makes sense

The outer exponent says to use 3^2 three times. That makes $2 + 2 + 2 = 6$ factors of 3. Multiplying the exponents is the shortcut for counting those six factors.

Question 10

Question

Simplify $(2 \cdot 5)^3$.
Checked answer: 1000

What the question is asking

This question asks you to find the number made by multiplying 2 and 5, then using that result three times as a factor.

Important words

- An *operation* is a math action. Add joins amounts. Subtract takes away or finds a difference. Multiply makes equal groups. Divide splits into equal groups.
- An *expression* is a math phrase made of numbers and operation signs.
- To *simplify* means rewrite an expression in a shorter form with the same value.
- To *evaluate* means find the number value of an expression.
- A *product* is the answer to multiplication.
- A *factor* is a number being multiplied.

- *Parentheses* are curved marks that group work together.
- The *base* is the number used as a repeated factor. Here the grouped base is $(2 \cdot 5)$.
- An *exponent* is the small raised number that tells how many times to use the base as a factor.
- A *power* is a base and its exponent together, such as $(2 \cdot 5)^3$.
- To *expand* a power means write all of its repeated factors.
- The *power-of-a-product law* says $(ab)^n = a^n b^n$.
- *PEMDAS* gives the order: Parentheses, Exponents, Multiply and Divide from left to right, then Add and Subtract from left to right.

Work the problem one tiny step at a time

1. PEMDAS letter P says to work inside the parentheses first.
2. Multiply inside the parentheses: $2 \cdot 5 = 10$.
3. Rewrite the expression: $(2 \cdot 5)^3 = 10^3$.
4. The exponent 3 means use 10 as a factor three times.
5. Expand: $10^3 = 10 \times 10 \times 10$.
6. Multiply the first two factors: $10 \times 10 = 100$.
7. Multiply by the last factor: $100 \times 10 = 1000$.
8. Check with the power-of-a-product law: $(2 \cdot 5)^3 = 2^3 \cdot 5^3$.
9. Evaluate: $2^3 = 2 \times 2 \times 2 = 8$.
10. Evaluate: $5^3 = 5 \times 5 \times 5 = 125$.
11. Multiply to check: $8 \times 125 = 1000$.

Final answer: 1000

Why this makes sense

The grouped base is $2 \cdot 5 = 10$. Using 10 three times gives 1000. Distributing the exponent to both factors also gives $8 \cdot 125 = 1000$.

Question 11

Question

Simplify $6^{-2} + \left(\frac{8}{4}\right)^3$.

Checked answer: $8\frac{1}{36}$

What the question is asking

This question asks you to find the number made by adding the two parts of the expression.

Important words

- An *operation* is a math action. Add joins amounts. Subtract takes away or finds a difference. Multiply makes equal groups. Divide splits into equal groups.
- An *expression* is a math phrase made of numbers and operation signs.
- To *simplify* means rewrite an expression in a shorter form with the same value.
- To *evaluate* means find the number value of an expression.
- *Nonzero* means not equal to zero.
- A *fraction* has a top number and a nonzero bottom number and names equal parts of a whole.
- A *whole number* has no fractional part, such as 8.
- The *base* is the number used as a repeated factor.
- An *exponent* is the small raised number that tells how many times to use the base as a factor.
- A *power* is a base and its exponent together, such as 6^2.
- A *reciprocal* is found by flipping a nonzero fraction. The reciprocal of $\frac{36}{1}$ is $\frac{1}{36}$.
- A *negative exponent* tells you to use the reciprocal of the matching positive power: $a^{-n} = \frac{1}{a^n}$.
- A *quotient* is the answer to division.
- A *proper fraction* has a top number smaller than its bottom number, such as $\frac{1}{36}$.
- An *improper fraction* has a top number at least as large as its bottom number, such as $\frac{289}{36}$.
- A *mixed number* has a whole number and a proper fraction.
- *PEMDAS* gives the order: Parentheses, Exponents, Multiply and Divide from left to right, then Add and Subtract from left to right.

Work the problem one tiny step at a time

1. PEMDAS letter P says to work inside the parentheses first.
2. Divide inside the parentheses: $8 \div 4 = 2$.
3. Rewrite that power: $\left(\frac{8}{4}\right)^3 = 2^3$.
4. PEMDAS letter E says to work with exponents next.
5. Rewrite the negative exponent: $6^{-2} = \frac{1}{6^2}$.
6. Evaluate 6^2: $6 \times 6 = 36$.

7. Therefore $6^{-2} = \dfrac{1}{36}$.
8. Evaluate the other power: $2^3 = 2 \times 2 \times 2$.
9. Multiply: $2 \times 2 = 4$.
10. Multiply: $4 \times 2 = 8$.
11. The expression is now $\dfrac{1}{36} + 8$.
12. Write the whole number first: $8 + \dfrac{1}{36}$.
13. Combine the whole number and proper fraction: $8 + \dfrac{1}{36} = 8\frac{1}{36}$.
14. Check as an improper fraction: $8 = \dfrac{288}{36}$, and $\dfrac{288}{36} + \dfrac{1}{36} = \dfrac{289}{36} = 8\frac{1}{36}$.

Final answer: $8\frac{1}{36}$

Why this makes sense

The negative power is a small positive amount, one thirty-sixth. The other power equals 8. Their sum must be a little more than 8, and $8\frac{1}{36}$ is a little more than 8.

Question 12

Question

Simplify $3^4 \cdot 3^2$.
Checked answer: 729

What the question is asking

This question asks you to find the number represented by multiplying two powers of 3.

Important words

- An *operation* is a math action. Add joins amounts. Subtract takes away or finds a difference. Multiply makes equal groups. Divide splits into equal groups.
- An *expression* is a math phrase made of numbers and operation signs.
- To *simplify* means rewrite an expression in a shorter form with the same value.
- A *factor* is a number being multiplied.
- The *base* is the number used as a repeated factor. Both powers have base 3.
- An *exponent* is the small raised number that tells how many times to use the base as a factor.
- A *power* is a base and its exponent together, such as 3^4.
- A *product* is the answer to multiplication.
- The *product-of-powers law* says to keep a matching base and add the exponents: $a^m \cdot a^n = a^{m+n}$.

- To *expand* a power means write all of its repeated factors.

Work the problem one tiny step at a time

1. Look at 3^4. Its base is 3.
2. Look at 3^2. Its base is also 3.
3. The bases match, so use the product-of-powers law.
4. Keep the base 3.
5. Add the exponents: $4 + 2 = 6$.
6. Rewrite: $3^4 \cdot 3^2 = 3^6$.
7. Expand: $3^6 = 3 \times 3 \times 3 \times 3 \times 3 \times 3$.
8. Multiply: $3 \times 3 = 9$.
9. Multiply: $9 \times 3 = 27$.
10. Multiply: $27 \times 3 = 81$.
11. Multiply: $81 \times 3 = 243$.
12. Multiply: $243 \times 3 = 729$.

Final answer: 729

Why this makes sense

Four factors of 3 and two more factors of 3 make six factors of 3. Expanding the six factors confirms that the value is 729.